



Naresh Modina

Machine Learning Engineer | AI/ML Researcher

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SKILLS

Programming	Python (advanced), HTML/CSS (Intermediate).
ML / DL	PyTorch (advanced), TensorFlow(Intermediate), Scikit-learn(advanced), NumPy / Pandas (advanced).
LLMs / NLP	Hugging Face Transformers, Datasets, SFT, PEFT / LoRA.
Reinforcement Learning	DQN, PPO, Actor-critic, Gymnasium, World Models.
MLOps	Docker, Git, torchrun (multi-GPU DDP), distributed training, Conda, SQL.
Soft Skills	Technical writing, presentation, team coordination, research grant proposals.

CURRENT POSITION

Applied AI Researcher 2023 – 2026

Conservatoire National des Arts et Métiers (Cnam), Paris, France.

RESEARCH EXPERIENCE

Projects

Tele-SLMs — Open Source Project 2026

- Explores whether sub-2B parameter SLMs can be specialized for 3GPP and ETSI telecommunications standards — accurately and with low retraining overhead. Two-stage pipeline: full fine-tuning on 1.87B tokens from 38,302 standards documents (15,054 3GPP Rel-8–Rel-19 + 23,248 ETSI), followed by LoRA instruction fine-tuning on Alpaca.
- Evaluated on Tele-Eval (10,000 standards-derived examples): SmoLLM-TS-135M-it achieves 31.5% Ans-PPL reduction and SemScore 0.6183 → 0.6504 vs baseline; SmoLLM-TS-360M-it achieves 20.6% Ans-PPL reduction and SemScore 0.6216 → 0.6572. Four models published on [HuggingFace](#) (2 base, 2 instruction-tuned). Multi-GPU training via PyTorch DDP on 3× L40S (48GB).
- Curated TeleSpec-Data — a training-ready telecom NLP dataset of 38,302 standards documents, available on [HuggingFace](#).

NEXASPHERE — European Project 2025 – 2027

Development of monitoring and state assessment solutions for 3D networks integrating terrestrial and non-terrestrial (NTN) components towards 6G. Design of DRL-based solutions for automatic anomaly resolution via dynamic network reconfiguration. Partners include SAFRAN, HPE, OHB, HellaSat, Stellantis.

INFLUENCE — BPIFrance Project 2022 – 2026

Design of the ARM (Autonomous Reconfiguration–Intent Management) framework to resolve performance anomalies in 5G/xG systems through automated, intent-based recovery. Used DRL for intelligent container sizing and VNF resource management. Filed a patent in collaboration with Orange research team. Partners include Orange and Nokia.

MAESTRO5G — ANR Project 2019 – 2022

Optimization of IoT data orchestration through Age of Information (AoI)-based pricing and Markov Decision Processes (MDP), reducing operational costs and designing 5G slicing frameworks with multi-level equity. Resulted in publications in IEEE TMC, ICC, and ITC. Partners include Orange, Nokia, Inria, CentraleSupélec, IPP.

Supervision & Dataset Leadership

5G3E Testbed 2023 – 2026

Led the development of an end-to-end 5G emulation testbed and directed the creation, curation, and publication of the [5G3E open-source dataset](#) for AI/ML tasks in 5G networks.

Funding Proposals

- [NEXASPHERE](#) (2024) — Contribution focused on AI-based resource management.
- iLab (2026) — Drafted a competitive research and innovation proposal outlining objectives, technical methodology, impact, and implementation roadmap.
- Pionnier de l'IA (2026) — Developed a comprehensive AI-focused funding proposal detailing innovative contributions, research strategy, and expected scientific and societal impact.

EDUCATION

PhD in Computer Science 2019 – 2022

Avignon Université, France — Thesis: Resource allocation in 5G wireless networks

Master of Telecommunication Engineering 2015 – 2019

Politecnico di Milano, Italy

TEACHING

AI4CI — European Master Programme 2024 – 2026

Teaching [Artificial Intelligence and Machine Learning for Connected Systems](#), demonstrating strong foundations across ML, DL, and RL applied to network systems.

PUBLICATIONS & PATENT

- Patent: Procédés de détection et de gestion d'anomalies impactant un environnement informatique ([FR3164046](#), 2024) — filed in collaboration with Orange.
- 4 peer-reviewed publications in IEEE TMC, ITC, and ICC on next-generation wireless networks. Full list available on [Google Scholar](#).

LANGUAGES

English (Professional fluency) | **French** (A2) | **Hindi** (Professional fluency) | **Telugu** (Native)